

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

BUILDING TECHNOLOGY AND DESIGN
PAPER 2 Construction Engineering

6003/02

3 hours

SPECIMEN PAPER

Additional materials:

- A3 drawing paper,
- Drawing instruments,
- Scientific calculator,

INSTRUCTIONS TO CANDIDATES

SECTION A [20 MARKS]

- Answer **all** questions on the spaces provided in the question paper.
- You are advised to spend not more than **30 minutes** in this section.

SECTION B [50 MARKS]

- Answer **all** questions on the separate answer booklet provided.
- Attach the answer sheets securely to the question paper.
- You are advised to spend not more than **1 hour 30 minutes** in this section.

SECTION C [30 MARKS]

- Answer **two** questions only on separate answer booklet provided.
- Attach the answer sheets securely to the question paper.
- You are advised to spend **not more than 1 hour** in this section.

INFORMATION FOR CANDIDATES

Marks are given in brackets [] at the end of each question.

This specimen paper consists of 9 printed pages.

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Section A

*You are advised to spend **not more than 30 minutes** in this section.*

*Answer **all** the questions on the spaces provided on the question paper.*

- 1 (a) Justify the use of the following materials in the construction industry:

- (i) thatch on a lodge in a resort area and

[1]

- (ii) stainless steel on a urinary.

[1]

- (b) Figure 1(b) shows a staff extract.

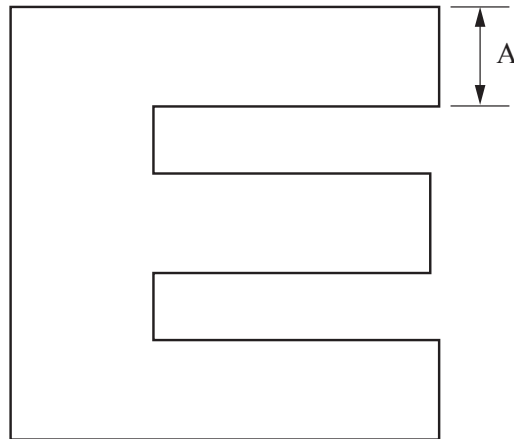


Fig. 1(b)

State the measurement labelled A.

[1]

- (c) State the units of measurement for the following building construction work when calculating quantities:

- (i) brickwork,

[1]

- (ii) concrete works.

[1]

- 2 Calculate the stiffness constant of a cantilever beam using the following specifications:

Force = **100 N**

Deflection = **0.03 M**

[5]

- 3 (a) (i) State **one** importance of building bye-laws in Construction Engineering.

[1]

- (ii) Justify the need for inspection of a ring beam by a structural engineer.

[1]

- (b) (i) Mention any **one** document required for engineering tendering.

[1]

- (ii) Justify the presence of Engineers during tender selection.

[1]

- (c) State a method of **shoring** to resist vertical loads.

[1]

- 4 (a) Identify any **three** traditional architectural features found on modern structures.

- (b) List any **two** conditions necessary for the production of biogas.

For
Examiner's
Use

SECTION B

Answer *all* questions in this section.

- 5 (a) **Figure 5(a)** shows a bridge of weight **100 kN** which has a car of weight **12 kN** on the position.

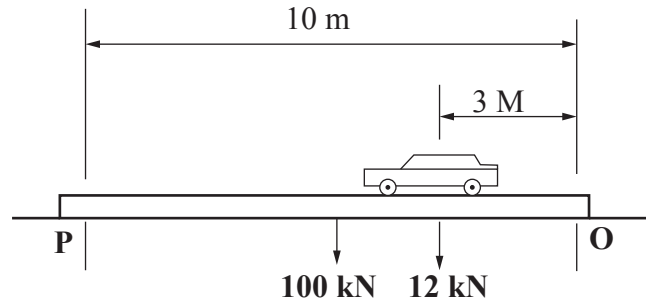


Fig. 5(a)

Determine the reactions at **P** and **Q**.

- (b) **Figure 5(b)** shows space diagram for a wall mounted crane.

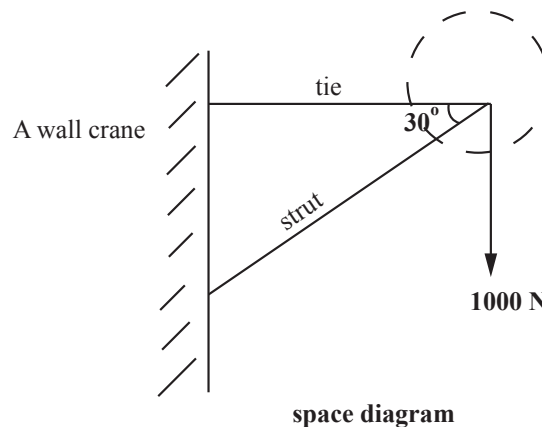


Fig. 5(b)

Determine the force created in the tie and strut by the **1 000 N** load. [5]

- 6 Illustrate how a new foundation concrete of a **230** wall is joined to an existing one. [5]
- 7 Illustrate the storage and distribution of hot water in a household. [5]

8 **Figure 8** shows a sketch floor and roof plan of a residential building.

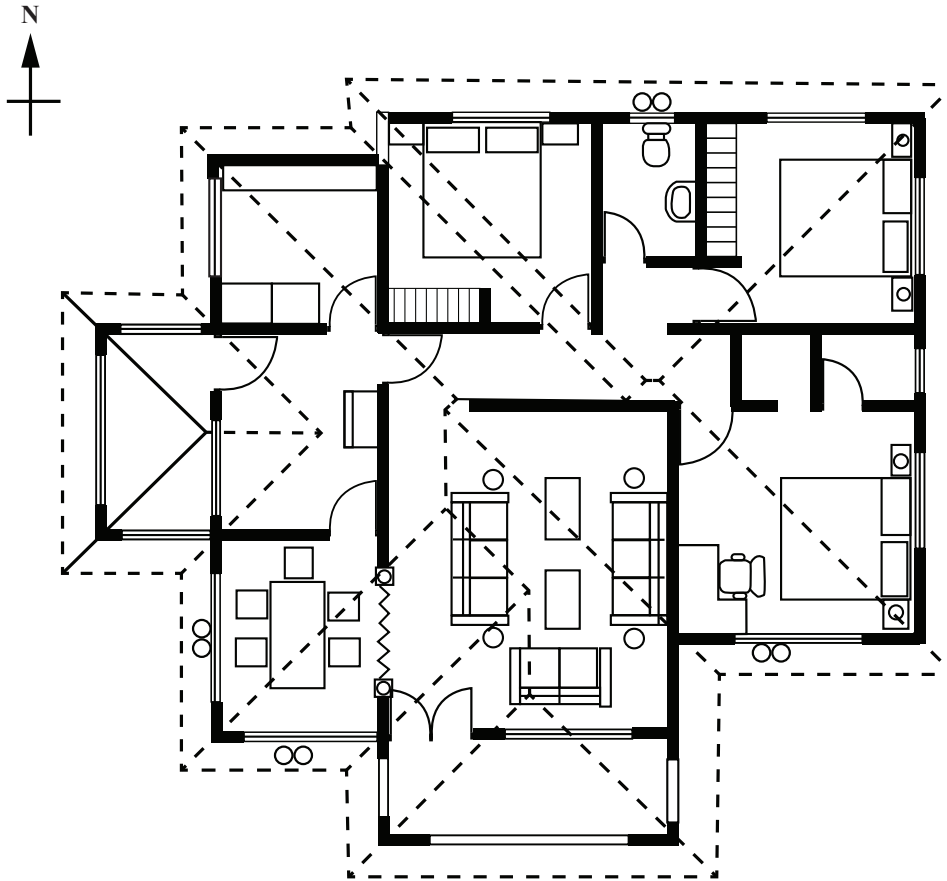


Fig. 8

Produce elevation **M**, incorporating features of ionic order.

[15]

9 (a) Illustrate compaction factor test for concrete.

[3]

(b) Label any 5 parts.

[4]

(c) Show dimensions.

[3]

10 (a) **Figure 10(a)** shows contour survey on a building site.

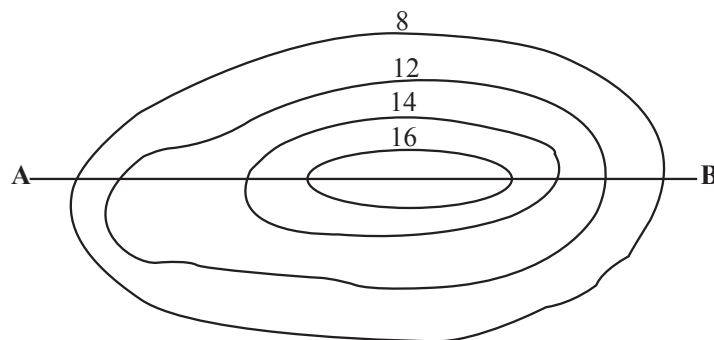


Fig. 10(a)

(i) Draw a cross-section of the terrain.

[2]

(ii) Show position of septic tank on the cross-section.

[1]

(b) Figure 10(b) shows a concrete footing.

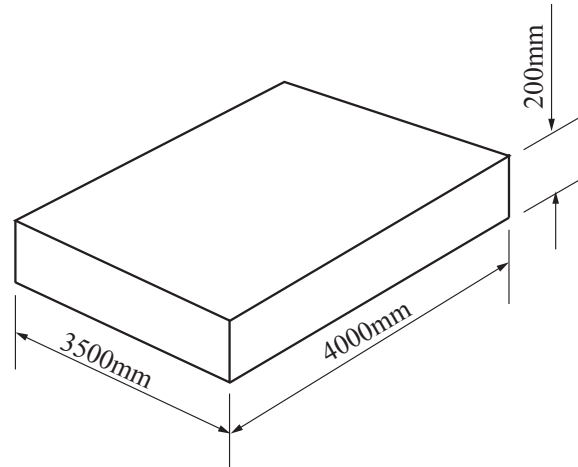


Fig. 10(b)

Specifications:

Concrete mix	-	1: 3: 6
Volume of one bag of cement	-	0,034

Calculate the number of cement bags.

[4]

(c) Figure 10(c) shows a pad foundation.

[3]

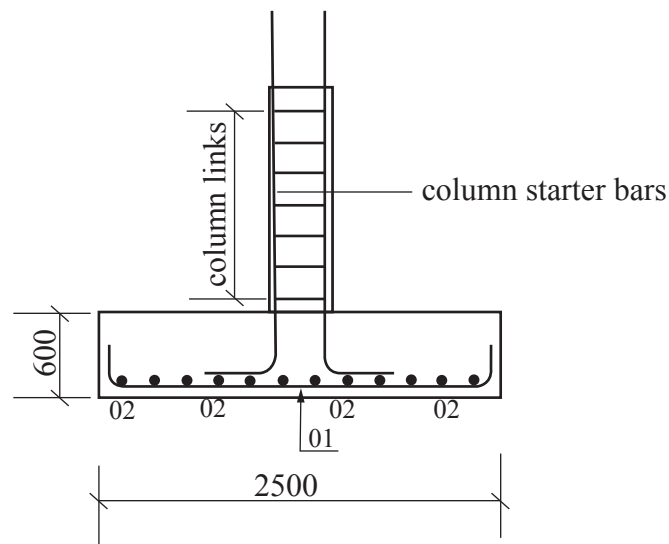


Fig. 10(c)

Calculate stress on the pad foundation.

[3]

SECTION C [30 Marks]

Answer **two** questions only.

- 11 (a)** Figure 11(b) shows a beam supported at two points.

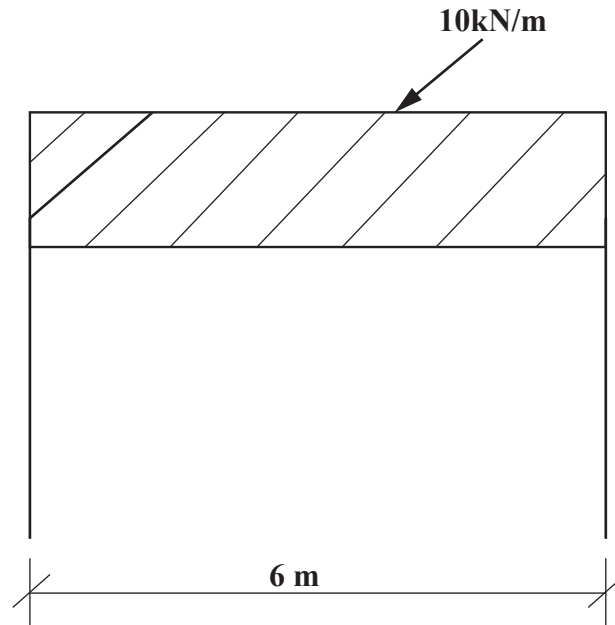


Fig. 11(a)

Calculate

- (i) shear force and [5]
- (ii) bending moment. [5]

- (b)** Table 11(b) shows the results of a test on the compaction factor of concrete.

Table 11(b) showing compaction factor results.

Factor	Partially compacted concrete	Fully compacted concrete
Weight (g)	20.5 kg	24.5 kg

Calculate compaction factor. [5]

- 12** **(a)** Design and draw a longitudinal cross-section of a straight flight staircase. [10]
- (b)** Justify the functions of the following materials in the construction of a staircase:
- (i)** cement, [1]
- (ii)** water, [1]
- (iii)** river sand, [1]
- (iv)** quarry stones and [1]
- (v)** steel rods. [1]
- 13** **(a)** **(i)** Design a **10 m** roof truss to support clay tiles. [7]
- (ii)** Show on the diagram forces acting on the truss members. [3]
- (b)** **(i)** Calculate the lateral load of a wall between buttress on a retaining wall using the following information:
- height - **3 m**
- lateral soil pressure - **6 kN/m²**
- spacing - **3 m**
- (ii)** List any **two** properties of materials for the construction of the buttress. [2]